GRADE 12 / Geography / English

Read a river’s long profile

01 Understand

Learning goals: Distinguish erosion, transport and deposition • Interpret a long profile • Connect gradient with channel processes

A river’s long profile shows elevation along its course. It commonly becomes gentler downstream, although local geology and disturbances create exceptions.

Erosion removes material, transport moves it and deposition occurs when the river can no longer carry part of its load.

In a meander, faster flow commonly erodes the outer bank while slower flow deposits material on the inner bank. Distinguish this cross-sectional pattern from the river’s long profile.

Simplified downstream pattern

Section	Typical gradient	Common tendency
Upper course	steeper	vertical erosion
Middle course	moderate	transport and lateral erosion
Lower course	gentler	deposition where carrying capacity falls

02 Follow the reasoning

Explain why the outside and inside banks of a meander may change differently.

1. Flow is commonly faster and deeper near the outer bank, promoting erosion and a river cliff.

2. Slower flow near the inner bank encourages deposition and a slip-off slope.

3. Over time this can move the meander laterally. Real outcomes also depend on discharge, sediment and bank material.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Define deposition.

2. A river falls 120 m over 6 km. Calculate average gradient as a ratio in the same units.

3. Why does gentler gradient not always mean lower discharge?

04 Check and correct

1. The laying down of sediment when transport capacity is insufficient to carry that load.

2. 120 : 6 000 = 1 : 50.

3. Discharge depends on cross-sectional area and velocity; tributaries may increase volume downstream.

Common mistake to avoid

Convert kilometres to metres before forming the gradient ratio. A profile is not a plan view of a winding river.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Geomorphology: fluvial processes; river profiles

DBE CAPS, Grades 10–12: Geography. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID geo-12.